



Simultaneous measurement of double surfaces of transparent lenses with phase measuring deflectometry

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ABSTRACT

The phase measuring deflectometry is a promising method for the high precision measurement of optical freeform surfaces. However, it fails to measure the double surfaces of transparent lenses, because the fringes reflected from both the upper and the lower surfaces are superimposed in the captured images. An effective method is proposed to solve this problem by displaying a set of fringe patterns with different frequencies and retrieving the fringe phases associated with both surfaces. The form of the lower surface is reconstructed using a hybrid reflection-refraction model. Experimental results demonstrate that the proposed method can reliably reconstruct both surfaces of a lens, making deflectometry capable of measuring transparent optics with high accuracy.

1. Introduction

The form quality is of great significance to optical lenses, which determines the opto-electronic performances of the major facilities and precision instruments in the fields of astronomical observation, medical diagnosis etc. With the increasing requirement on the imaging functionalities of the optical systems, the shapes of the functional surfaces are becoming increasingly complex, from simple planar/spherical surfaces to aspherical/freeform surfaces. The commonly used interferometry has a limited measuring range of several microns, thus it is not suitable for measuring complex surfaces, especially at the early stage of optical manufacturing, when the form deviation with respect to the nominal shape may be up to a sub-millimeter level [1].

Phase measuring deflectometry (PMD) is an efficient, flexible and robust measuring technique for complex surfaces, which displays fringe patterns on a screen, captures the fringes reflected from the workpiece, identifies correspondence pairs according to the phases of camera and screen pixels, calculates the surface gradients from the angular bisectors, and then reconstructs the surface by numerical integration [2]. The dynamic range of PMD is 1000 times higher than the conventional interferometry [1,3]. As a consequence, PMD has a great potential in the precision measurement of optical components [4]. However, a lot of optical components are transparent and they work as refractive or hybrid reflective/refractive components. Then the captured patterns are actually the superposition of the fringes reflected from the upper and lower surfaces. This type of ‘ghost’ fringe pattern cannot be demodulated by using the conventional phase-shifting methods [5].

In the past decades, researchers proposed several methods to overcome the ‘ghost’ pattern problem. Transmission deflectometry captures the deformed patterns transmitting directly through the lens under test. Evidently only the thickness information can be obtained rather than the forms of the double surfaces. Thus, additional conditions should be provided to decouple the information. A choice is to assume one surface is planar, then the form of the other curved surface can be obtained subsequently [6]. However, the assumption of the planar surface cannot be generally met in practice. Using dual wavelength transmission can also provide extra conditions. Since the refractive indices vary at different wavelengths, the phase information can be decoupled by recording the deformed patterns [7]. However, the refractive indices of two wavelengths are not significantly different, thus the measurement stability is limited. The two surfaces can also be specified using two transmission-mediums [8]. The refractive indices of the two selected mediums are quite different, which leads to a variation of optical length and consequently allows to decouple the double surfaces successfully. But this method is inconvenient to be used due to the changing of the medium. More importantly, the main disadvantage of transmission deflectometry is its low measurement sensitivity compared with the reflection deflectometry [6].

As for the reflection deflectometry, several techniques have been developed to decouple the superposed patterns. Blackening the lower surface or illuminating the lens with ultraviolet light is an effective approach of inhibiting the reflection from the lower surface [9]. Unfortunately, blackening the surface might destroy the element and it is expensive to utilize ultraviolet light sources. Su et al separated the superposed fringes by displaying a narrow fringe band [10]. But this method

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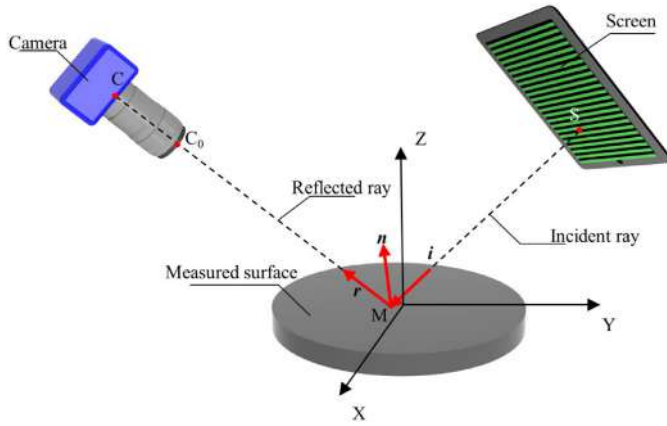


Fig. 1. Schematic of the deflectometric measurement.

is inefficient and there exists some ill-conditioned areas, where the two overlapped fringes cannot be decoupled. By line-shifting or gray coding method, Xu et al eliminated the signals reflected via the lower surface by thresholding [11]. However, setting a threshold may not be easy if the measured object has high transparency. The superposed fringes can also be decoupled by numerical algorithms. A frequency-shifting method using multi-frequency fringes and the Fourier-transform was developed, which considers the modulation transfer function as linear [12]. However, this approximation only can be used for planar elements, otherwise non-negligible measurement error will be resulted. Besides, the method requires about 2000 patterns to be captured, which greatly reduces the measuring efficiency. Fringes with two frequencies can be used to solve the phases of the double surfaces [5]. But due to the introduction of the flood-fill operation, the robustness of this algorithm is limited for practical applications. Furthermore, a drawback of the aforementioned methods is that they can only measure a single surface at one time. However, it is usually necessary to measure the double-sided surfaces of transparent elements [8].

As a consequence, a hybrid reflection-refraction deflectometry model is proposed to decouple the superposed fringes and reconstruct the two surfaces simultaneously without extra equipment. The rest of the paper is organized as follows. Section 2 demonstrates the absolute phase retrieval algorithm. Section 3 presents the hybrid reflection-refraction deflectometry model and the measurement procedure. Experimental verification is conducted in Session 4 and finally the paper is summarized in Section 5.

2. Phase decoupling from superposed fringes

2.1. Principle of reflection deflectometry

Fig. 1 schematically shows how x-gradients and y-gradients are measured on an optical surface. On one side of the system, a screen displaying fringe pattern faces to the element under test. On the other side, the camera captures the image of the screen reflected from the surface. Due to the pin-hole camera model, only the rays passing through the camera's optical center C_0 are considered. Thus, suppose the ray from a camera pixel point C passing through C_0 shoots at a point M on the mirror, and then the ray is reflected to a screen pixel S . According to the law of reflection, the angular bisector of the angle $\angle CMS$ is along the direction of the normal vector $\mathbf{n}$ at M . Hence, $\mathbf{n}$ can be expressed as

$$\mathbf{n} = [n_x \ n_y \ n_z]^T = \mathbf{r} - \mathbf{i} = [r_x - i_x \ r_y - i_y \ r_z - i_z]^T \quad (1)$$

Then, the x-gradient and the y-gradient at M , denoted with g_x and g_y , respectively, can be calculated as [3]

$$\begin{cases} g_x = -\frac{n_x}{n_z} = -\frac{r_x - i_x}{r_z - i_z} = -\frac{\frac{x_C - x_M}{d_{C2M}} + \frac{x_S - x_M}{d_{S2M}}}{\frac{z_C - x_M}{d_{C2M}} + \frac{z_S - x_M}{d_{S2M}}} \\ g_y = -\frac{n_y}{n_z} = -\frac{r_y - i_y}{r_z - i_z} = -\frac{\frac{y_C - y_M}{d_{C2M}} + \frac{y_S - y_M}{d_{S2M}}}{\frac{z_C - x_M}{d_{C2M}} + \frac{z_S - x_M}{d_{S2M}}} \end{cases} \quad (2)$$

where x_M , y_M and z_M are the estimated x, y and z coordinates of the point M , x_C , y_C and z_C are the coordinates of the camera pixel, x_S , y_S and z_S are the coordinates of the corresponding screen pixel, d_{C2M} is the distance from M to the camera pixel C , and d_{S2M} is the distance from M to the screen pixel S . Subsequently, the surface form can be reconstructed from the gradients [13]. To determine the correspondence relationship between the screen and camera pixels, a camera pixel and a screen pixel occupying the same phases are regarded as a pair. Therefore, phase retrieval from the captured fringe patterns is critical to the measuring accuracy of PMD.

2.2. Phase decoupling of superposed fringe patterns

It is known that the fringe pattern reflected by an optical surface can be described as

$$I = A + B \cos \mu \quad (3)$$

where I , A , B , and μ are the captured intensity, the average intensity, the modulation coefficient, and the phase value, respectively.

However, in contrast to the measurement of an opaque mirror, both sides of a transparent lens can reflect fringe patterns into the camera. Therefore, the resulting fringe intensity becomes

$$I = A + B_1 \cos \mu_1 + B_2 \cos \mu_2 \quad (4)$$

where B_1 and B_2 are the modulation coefficients associated with the upper and lower surfaces, respectively, and μ_1 and μ_2 are the fringe phases associated with the two surfaces, respectively. It is worth mentioning that the multi-reflections between the two surfaces are ignored in this equation. Since the reflectivity of the glass is about 4%, the relative intensities of the multi-reflected rays are only 1/1000 with respect to the directly reflected rays.

In order to attain the upper surface form of the measured lens, it is necessary to ascertain the corresponding phase value μ_1 . Considering Eq. (4) as a function associated with five variables A , B_1 , B_2 , μ_1 and μ_2 , a set of fringe patterns with different frequencies f , $k_1 f$, $k_2 f$, ..., and $k_m f$ is displayed without phase shifting to provide sufficient constraints for phase retrieval. When the frequency of the fringe pattern changes from f to $k_w f$, the resulting phase will be k_w times the original value. As a result, the intensity I_w can be written as

$$I_w = A + B_1 \cos(k_w \mu_1) + B_2 \cos(k_w \mu_2) \quad (5)$$

Henceforth, the unknown variable A can be cancelled out by subtracting Eq. (4) from Eq. (5). Based on the relation $\cos[k_w \mu] = \cos[(k_w - 1)\mu] \cos \mu - \sin[(k_w - 1)\mu] \sin \mu$, the following equation can be obtained

$$R_{w1} Y_1 + R_{w2} Y_2 + R_{w3} Y_3 + R_{w4} Y_4 = I_w - I_0 \quad (6)$$

where

$$\begin{aligned} R_{w1} &= \cos[(k_w - 1)\mu_1] - 1, Y_1 = B_1 \cos \mu_1, \\ R_{w2} &= -\sin[(k_w - 1)\mu_1], Y_2 = B_1 \sin \mu_1, \\ R_{w3} &= \cos[(k_w - 1)\mu_2] - 1, Y_3 = B_2 \cos \mu_2, \\ R_{w4} &= -\sin[(k_w - 1)\mu_2], Y_4 = B_2 \sin \mu_2 \text{ and } k_0 = 1 \text{ is the coefficient associated with } I_0. \end{aligned}$$

Y_1 , Y_2 , Y_3 , and Y_4 are the intermediate variables defined to facilitate solving the phases. Evidently, at least four independent equations

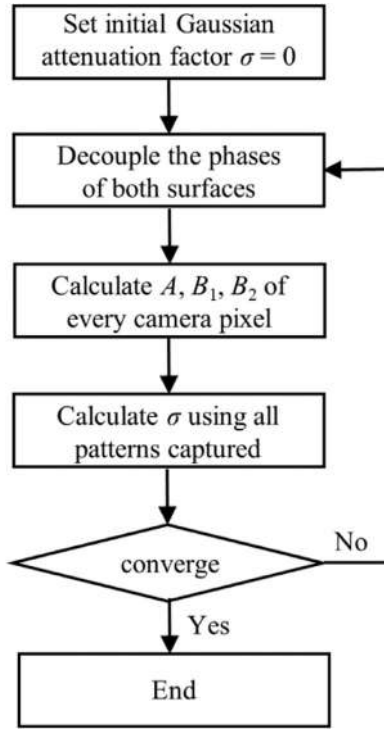


Fig. 2. Flowchart of retrieving phase of double surfaces.

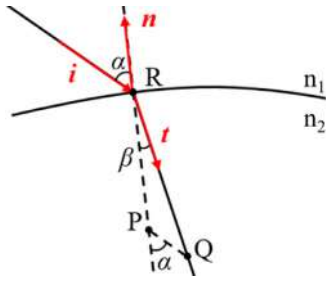


Fig. 3. Schematic diagram of refraction path.

are required to solve the four unknown variables B_1 , B_2 , μ_1 and μ_2 numerically. Once working out Y_1 , Y_2 , Y_3 , and Y_4 from the patterns, μ_1 and μ_2 can be obtained as

$$\mu_1(x, y) = \tan^{-1}\left(\frac{Y_2}{Y_1}\right), \quad \mu_2(x, y) = \tan^{-1}\left(\frac{Y_4}{Y_3}\right) \quad (7)$$

It is obvious that solving μ_1 and μ_2 is equivalent to calculating Y_1 , Y_2 , Y_3 , and Y_4 . It is worth mentioning that there are not any approximations in the derivation of these equations. In order to reduce the effect of the measurement errors by averaging, it is preferred to have more equations. Assuming that m patterns with different frequencies are captured, the system of equations can be written as

$$RY = T \quad (8)$$

$$\text{where } R = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ \vdots & \vdots & \vdots & \vdots \\ R_{m1} & R_{m2} & R_{m3} & R_{m4} \end{bmatrix}, \quad Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix}, \quad T = \begin{bmatrix} I_1 - I \\ I_2 - I \\ \vdots \\ I_m - I \end{bmatrix}$$

The elements in R and Y are defined as described in Eq. (6). Y can be solved with the least squares fitting

$$Y = (R^T R)^{-1} R^T T \quad (9)$$

Since R is determined by μ_1 and μ_2 , Y can be solved accordingly. In reverse, μ_1 and μ_2 can be recalculated from the obtained Y based

on Eq. (7). Henceforth, the solution of μ_1 and μ_2 and the calculation of Y are in fact mutually dependent. Therefore, these two sub-procedures can be calculated alternately until convergence.

- Step 1: Set the initial values of the phases by reverse ray tracing [14].
- Step 2: Calculate $Y^{(k)}$ with the current phases using Eq. (9).
- Step 3: Compute $\mu_1^{(k)}$ and $\mu_2^{(k)}$ with $Y^{(k)}$ using Eq. (7). Then unwrap $\mu_1^{(k)}$ and $\mu_2^{(k)}$ according to the initial phase values. It should be noted that discontinuous phase jumps may be present, since the initial fringe orders may not be exact.
- Step 4: Check the convergence of the solved phases $\mu_1^{(k)}$ and $\mu_2^{(k)}$ by $|\mu_1^{(k-1)} - \mu_1^{(k)}| + |\mu_2^{(k-1)} - \mu_2^{(k)}| < \epsilon$, where ϵ is a user-set threshold. If the condition is not met, change k to $k+1$ and return to step 2.

Because the process is implemented iteratively, the selection of the initial values is critical to guarantee correct convergence of the solution. Unlike the initial phases obtained according to the relative positions of the optical system specified in geometrical calibration [14], the initial modulation coefficients cannot be provided accurately because of the change of the surface slopes in different regions. Fortunately, the modulation coefficients are not utilized in this iterative process, which can effectively avoid false convergence due to the false setting of the initial modulation coefficients. In addition, in Eq. (8), the rank deficiency of the matrix R at some points will lead to false solutions. The phases associated with these points can be approximately obtained by interpolation based on the continuity of the phase maps of smooth surfaces. As a consequence, both the phase demodulation and phase unwrapping can be accomplished simultaneously.

In the previous model, the modulating coefficients B_1 and B_2 are set as constants at one pixel for different fringe frequencies. However, this does not hold true in theory. Since the stop of the camera is not an ideal point, the imaging process can be regarded as a convolution between the displayed fringe patterns and the point spread function of the deflectometric measurement system. Viewed from the frequency domain, the fringe amplitudes of high frequency components will decay due to the low-pass properties of the modulation transfer function of the imaging system. If a Gaussian low-pass model is adopted to describe the decay of the fringe amplitudes, Eq. (5) can be rewritten as

$$I_w = A + B_1 e^{-k_w^2 \sigma} \cos(k_w \mu_1) + B_2 e^{-k_w^2 \sigma} \cos(k_w \mu_2) \quad (10)$$

where σ is the decay factor. The variables are separated into groups and solved alternatively, as shown in Fig. 2. The initial Gaussian decay factor is set first as $\sigma = 0$ and the phases associated the double surfaces are solved with Eq. (9). Next, substitute the decay factor into Eq. (10) and calculate the coefficients A , B_1 and B_2 for every camera pixel. Then update σ for the whole images by the least squares fitting of Eq. (10). Subsequently the phases can be adjusted by using the new coefficients. The process above continues until σ converges or the fitting residual stagnates.

To make the fringe amplitudes stable, it is better to restrict the bandwidth of the fringes. Here we require the coefficients $\{k_w\}$, $w=1, 2, \dots, m$ to satisfy a condition $0.5 < k_w < 2$. On the other hand, if the bandwidth of $\{k_w\}$ is too narrow, the matrix R in Eq. (8) may be ill-conditioned. Therefore, the number of the displayed fringes should be more than 30 to guarantee the phases are distinct for different fringe frequencies, and the solution can be stably obtained.

3. Hybrid reflection-refraction deflectometry for the lower surface

Using the proposed phase retrieval method, the absolute phases of the upper and lower surfaces can be obtained. Thus, the form and location of the upper surface can be determined by the iterative SCOTS measurement procedure [13,15,16]. Then the lower surface can be calculated using the following method.

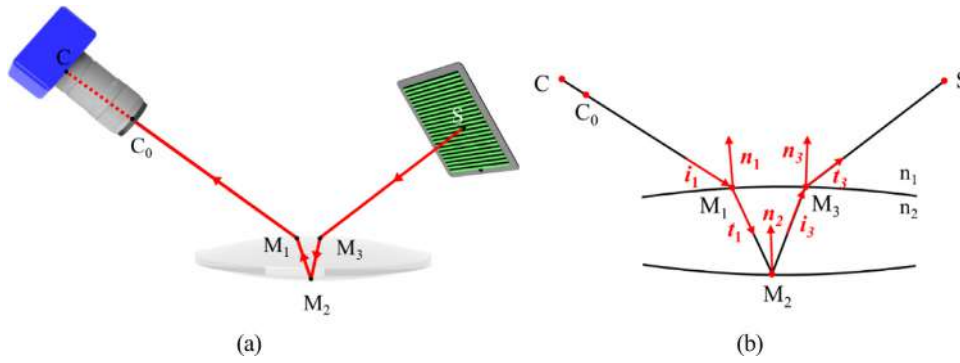


Fig. 4. Hybrid reflection-refraction model. (a) Light path associated with the lower surface; (b) two-dimensional simplified diagram.

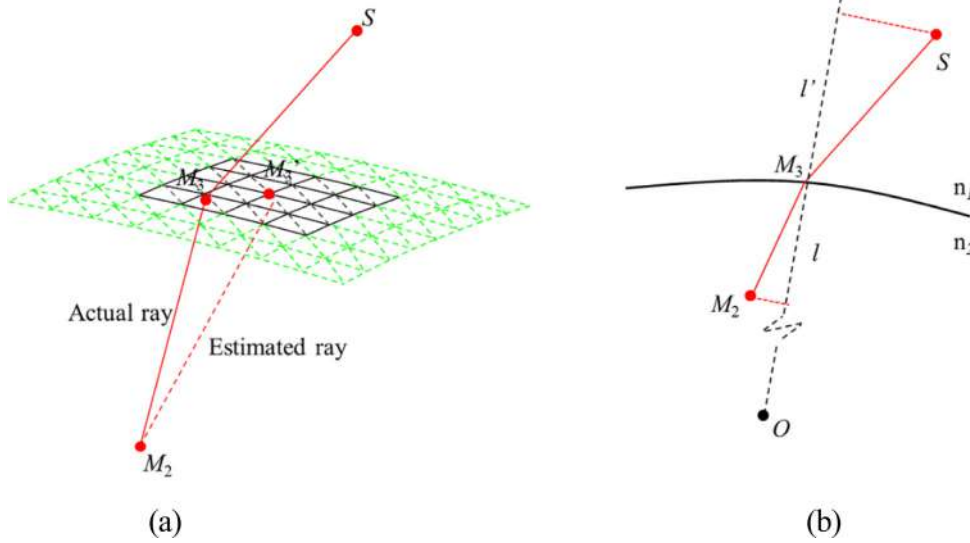


Fig. 5. Determination of M_3 from the upper surface. (a) Triangulation of the upper surface; (b) Light path of refraction.

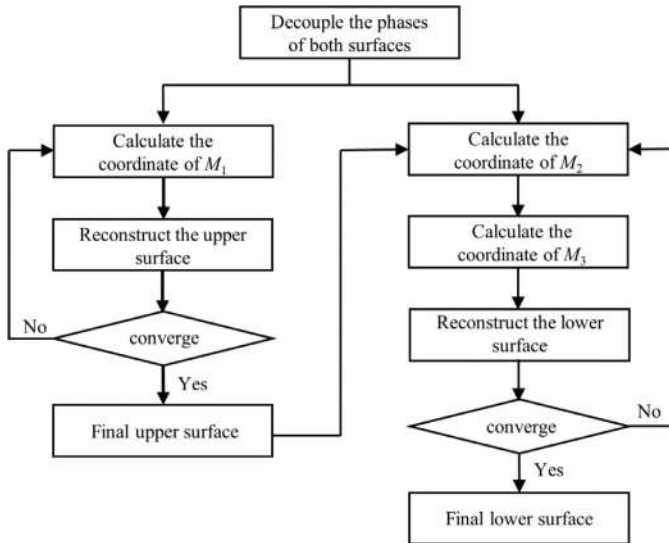


Fig. 6. Flowchart of the constructions of the both surfaces of a transparent lens.

3.1. Correction of rays in refractive propagation

Because refraction occurs when a ray enters the lens, the laws of refraction in the vector form is illustrated first. As the incident ray, the normal vector and the refractive ray are coplanar, a two-dimensional schematic is sufficient, as depicted in Fig. 3, where $\mathbf{i}$, $\mathbf{t}$ and $\mathbf{n}$ are all unit

vectors. α denotes the incident angle, β denotes the refractive angle, and n_1 and n_2 are the refractive indices of the two mediums, respectively. A triangle ΔRPQ is generated with $\overline{PQ}$ parallel to $\mathbf{i}$.

Based on the law of refraction, the following equation can be obtained

$$\frac{|RQ|}{|PQ|} = \frac{\sin \alpha}{\sin \beta} = \frac{n_2}{n_1} = n_{21} \quad (11)$$

where n_{21} denotes the ratio of n_2 and n_1 . Because $\overline{PQ}$, $\overline{RQ}$ and $\overline{PR}$ are along the directions of $\mathbf{i}$, $\mathbf{t}$ and $\mathbf{n}$, the following relationship can be obtained

$$\mathbf{n} = \lambda(n_{21}\mathbf{t} - \mathbf{i}) \quad (12)$$

where λ is a coefficient used to guarantee that $\mathbf{n}$ is a unit vector,

$$\lambda = \frac{\cos \alpha - \sqrt{\cos^2 \alpha + n_{21}^2 - 1}}{n_{21}^2 - 1} \quad (13)$$

where $\cos \alpha = \langle \mathbf{i}, \mathbf{n} \rangle$ represents the inner product of $\mathbf{i}$ and $\mathbf{n}$. It can be seen from Eq. (12) and Eq. (13) that the refractive propagation can be uniquely determined by $\mathbf{i}$, $\mathbf{n}$ and n_{21} .

The rays in the refractive propagation can be corrected as follows. Fig. 4(a) shows that the image of the screen is reflected by the lower surface and finally captured by the camera. A ray emitted from a screen pixel point S shoots at the camera pixel point C .

The reverse ray tracing process of this ray propagation is demonstrated in Fig. 4(b). A ray emitted from C passes through the camera center C_0 and intersects with the upper surface, lower surface, and upper surface at M_1 , M_2 and M_3 , respectively, and finally shoots at the screen pixel point S . $\mathbf{i}_1$, $\mathbf{n}_1$ and $\mathbf{t}_1$ denote the directions of the incident ray, normal and refractive ray at M_1 , respectively. $\mathbf{t}_1$, $\mathbf{n}_2$ and $\mathbf{i}_3$ are the

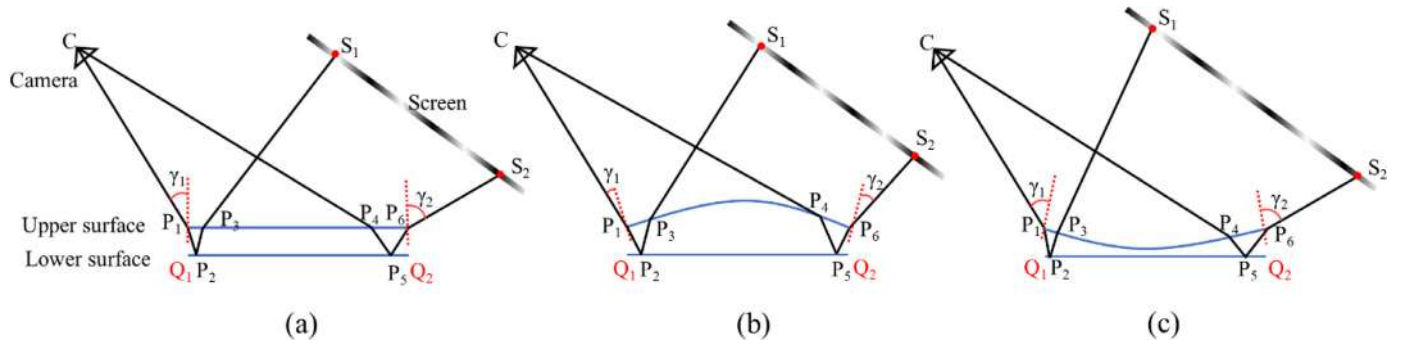


Fig. 7. Two-dimensional diagram of the measured areas of double surfaces. (a) flat upper surface; (b) convex upper surface; (c) concave upper surface.

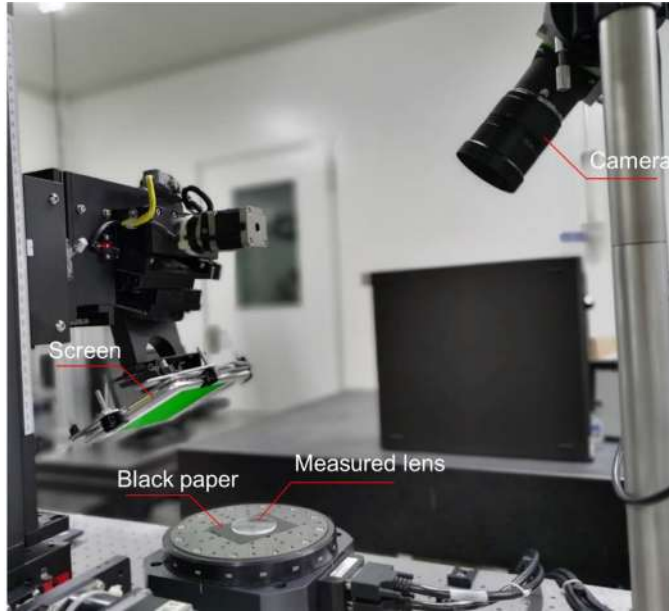


Fig. 8. The deflectometric measuring system.

corresponding directions at M_2 . i_3 , n_3 and t_3 represent the corresponding directions at M_3 . With the well-reconstructed upper surface, the position of M_1 and its normal vector n_1 are already known. Then t_1 can be calculated from i_1 and n_1 according to Eq. (12) and (13). The point M_2 can be specified subsequently with the provided current position and nominal model of the lower surface. Furthermore, the point M_3 can be determined provided with M_2 , S and the upper surface. Eq. (12) turns

out to be

$$v = w(n_{21}i_3 - t_3) \quad (14)$$

where v is the normal vector at M_3 calculated with the law of refraction and w is a normalization coefficient. However, the actual normal vector n_3 at M_3 can be worked out based on the reconstructed upper surface. Henceforth the deviation between v and n_3 reflects the error of the estimated position of M_3 , equivalently the position error of M_3 . Then the calculation of M_3 becomes

$$\min_{M_3} \|v - n_3\| \quad (15)$$

But the reconstructed upper surface is a set of discrete points, it is inconvenient to calculate the M_3 coordinate. Since the upper surface is smooth, it is reasonable to set the point closest to the estimated ray r_2 (also i_3) as the initial M_3 and denote it as M_3' , as illustrated in Fig. 5 (a). In order to obtain the correct coordinate of M_3 , a triangular mesh is constructed [17], as shown in Fig. 5 (a). The normal vector of the triangle is assigned to n_3 and the point coordinate in the triangle is applied for calculating v . Then traverse all the triangles within a proper range of neighborhood around M_3' as depicted with solid lines. In this way the optical solution of Eq. (15) can be obtained.

Furthermore, it is necessary to prove that the solution of M_3 is unique on the upper surface. As illustrated in Fig. 5(b), the ray emitted from M_2 passes through M_3 and shoots at S . If there exist multiple points satisfying Eq. (15), M_2 and S must have a conjugate object-image relationship

$$\frac{n_2}{l} - \frac{n_1}{l'} = \frac{n_1 - n_2}{r} \quad (16)$$

where r is the radius of curvature of the fitted sphere associated with the local area around M_3 , and l and l' are the object distance and image distance associated with M_2 and S , respectively. In fact, r is much greater than l , then it is ascertained that Eq. (16) cannot hold true. As a consequence, M_2 and S do not obey a conjugate relationship, implying that the solution of M_3 is unique.

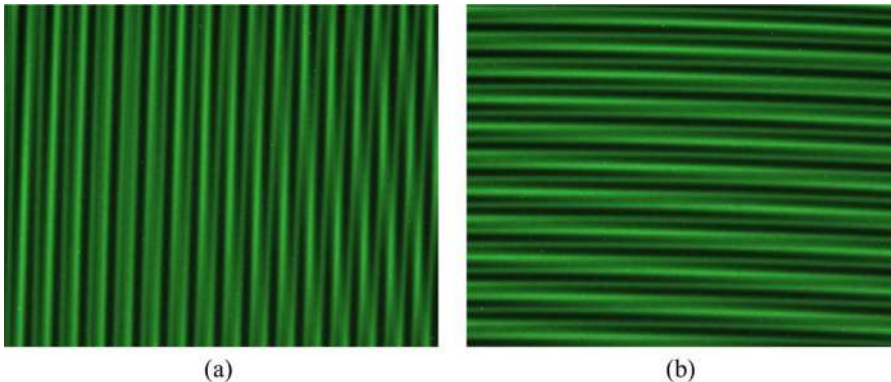


Fig. 9. The acquired coupled patterns of the upper and lower surfaces. (a) x fringe pattern; (b) y fringe pattern.

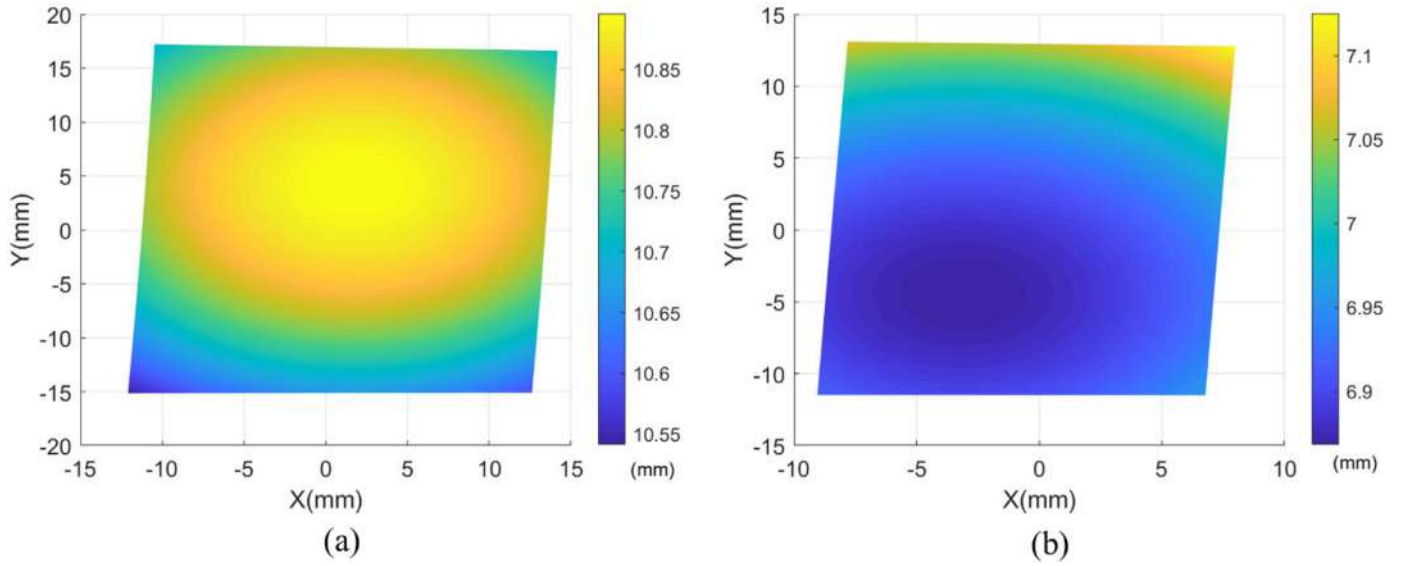


Fig. 10. The measured results of the lens. (a) Upper surface; (b) Lower surface.

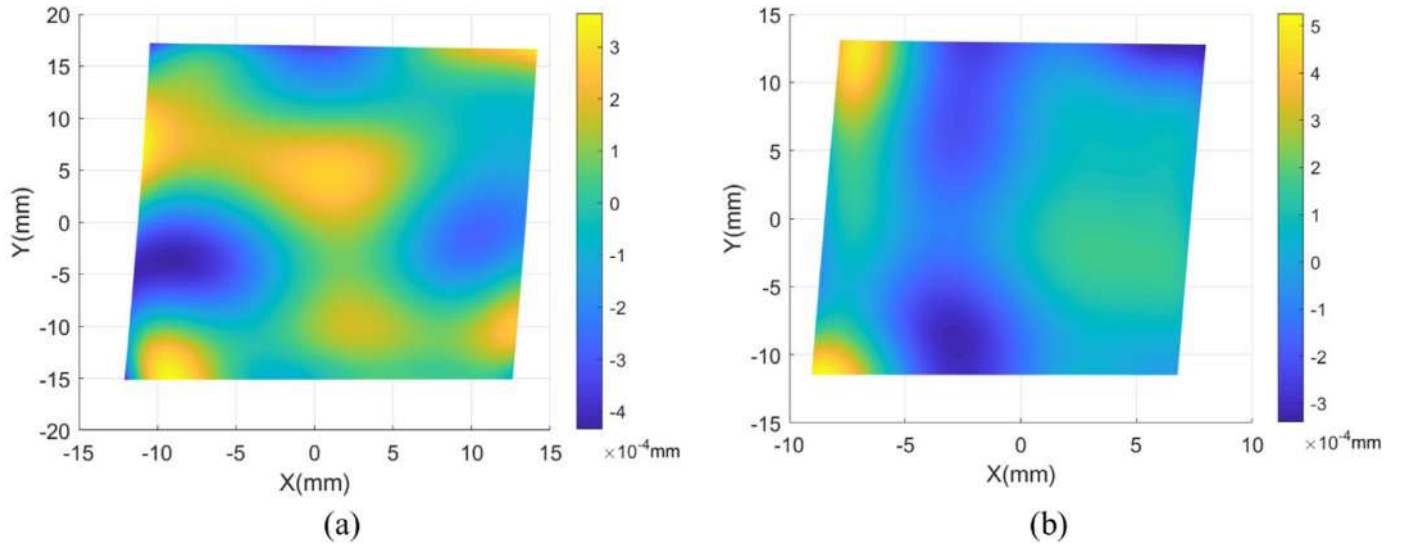


Fig. 11. The measurement deviations of both surfaces. (a) Upper surface; (b) Lower surface.

3.2. Reconstruction of the lower surface

Once M_3 is specified, the corresponding incident ray i_3 can be determined. The angular bisector between M_3M_2 and M_2M_1 is regarded as the normal vector at M_2 . Henceforth, the x -gradients and y -gradients of the lower surface can be obtained, and the surface form can be reconstructed subsequently. Similar to the SCOTS approach in the reflection deflectometry, the new reflecting point M_2 is again estimated as the intersections between the ray t_1 and the current lower surface, and the point M_3 is updated by solving Eq. (15). This process is iterated until convergence.

In order to clarify the reconstruction process of both surfaces of the lens, the whole procedure is represented in Fig. 6.

Since the gradients of the lower surface are calculated based on the form and location of the upper surface, the measured area of the lower surface will be a little smaller, as shown in Fig. 7. With the well-reconstructed upper surface P_1P_6 , the measurable area of the lower surface is P_2P_5 . Thus, Q_1P_2 and Q_2P_5 are the unmeasurable areas. Their sizes are related to the curvatures of the two surfaces and the thickness of between them.

4. Experimental verification

In order to quantitatively verify the performance of the proposed method, a deflectometric measurement system is established as presented in Fig. 8. The used camera is JAI SP-2000C-PMCL with the resolution of 5120×3840 pixels. An iPad mini2 with the resolution of 2048×1536 pixels plays as a screen. Since the camera is more sensitive to green light, green patterns with 90 fringes are displayed on the screen to avoid chromatic aberrations. A standard K9 biconvex lens with a radius of curvature of both sides of 771.99 mm and an aperture of 50.8 mm is adopted as a lens under test. A piece of rough black paper is placed directly under the lens aiming to prevent stray light. The relative positions of the whole measuring system are calibrated using the method described in [18].

In the experiment, the frequencies of the displayed fringe patterns are 1.031, 1.073, 1.113, 1.139, 1.163, 1.191, 1.211 times the original pattern frequency, respectively. The fringes are displayed sequentially along the x -direction and y -direction. Fig. 9(a) and (b) illustrate a part of the captured images of the fringe patterns.

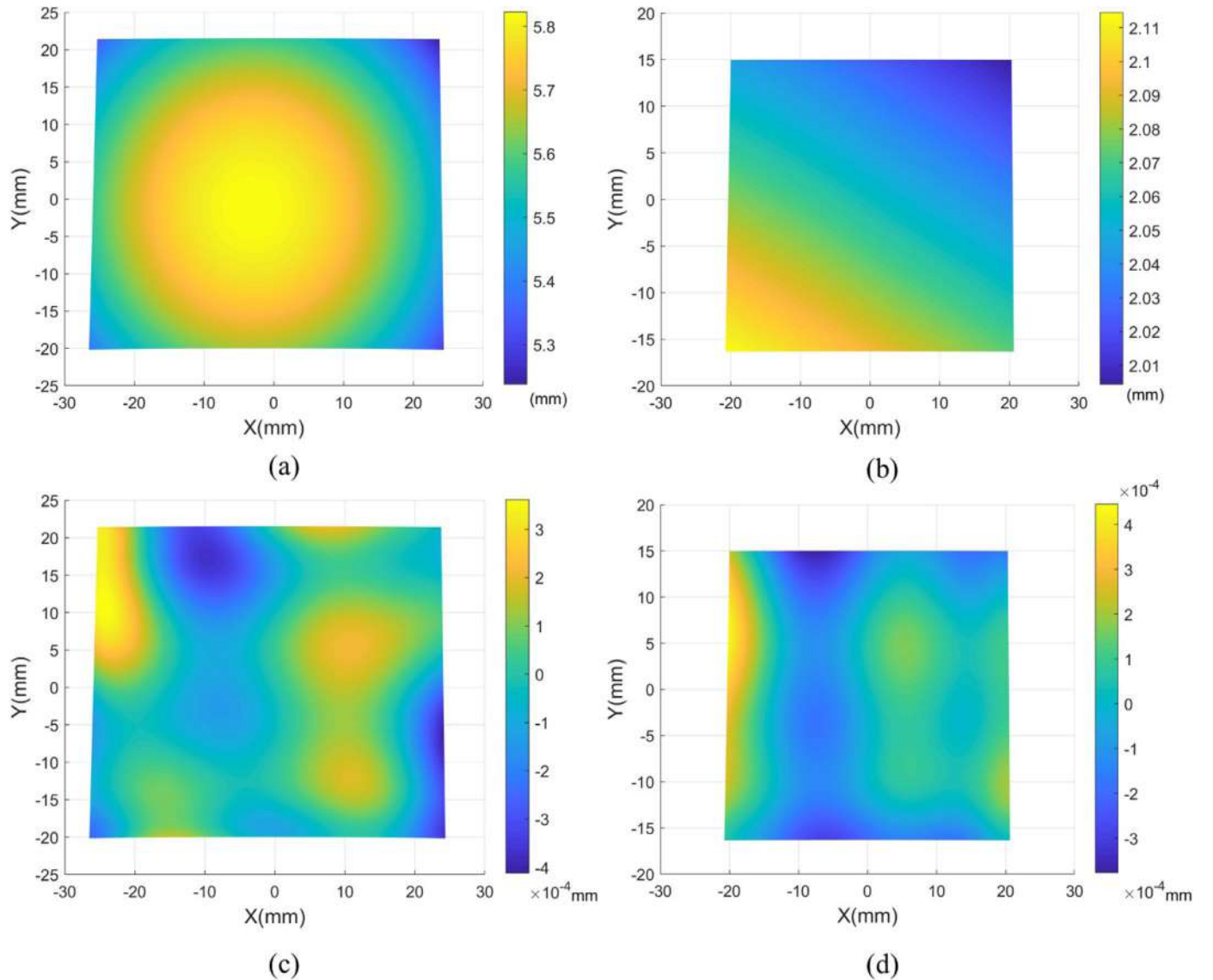


Fig. 12. The measurement result of plano-convex lens. (a) Form of the upper surface; (b) Form of the lower surface; (c) Deviation of the upper surface; (d) Deviation of the lower surface.

Then the initial phases of the two surfaces can be approximately obtained via the reverse ray tracing method based on the calibrated system [14]. With the phase retrieval approach presented in Section 2.2, the accurate absolute phases of the upper and lower surfaces can be calculated without phase unwrapping. It worth mentioning that if considering the decay of the fringe amplitudes in the captured images, the obtained decay factor σ is 0.064. The resulting phase deviation is about 0.01 rad, implying that the decay of amplitudes associated with different fringe frequencies can be ignored. Henceforth the phases can be directly solved from Eq. (5).

Then the correspondence relationship between the camera pixels and the screen pixels is identified according to their phases. The x -gradients and y -gradients of the upper surface are obtained based on Eq. (2) and subsequently the upper surface is reconstructed as illustrated in Fig. 10(a). With the well reconstructed upper surface, the lower surface is obtained by the proposed method presented in Section 3, as depicted in Fig. 10(b).

To verify the accuracy of the proposed method, the measurement results are compared with the surface profiles provided by the manufacturer, as shown in Fig. 11. The residual RMS (Root-Mean-Squares) and PV (Peak-To-Valley) errors of the upper surfaces turn out to be 158.7

nm and 799.3 nm, while those of the lower surface are 151.8 nm and 865.5 nm, respectively. It is worth mentioning that the deviations are mainly caused by the low reflectivity of the glass, yielding a low signal to noise ratio of the captured images.

Another standard K9 plano-convex lens is measured demonstrate the general applicability of the proposed method. The convex surface has a radius of curvature of 1033.65 mm and an aperture of 80.0 mm. The measurement results and the deviations with respect to the figures provided by the manufacturer are presented in Fig. 12. The RMS and PV errors of the upper surfaces turn out to be 131.6 nm and 772.5 nm, while those of the lower surface are 138.6 nm and 821.9 nm, respectively. As a consequence, this method can be generally applied to measuring transparent lenses with different shapes.

5. Conclusion

This paper presents a phase decoupling algorithm for the overlapped fringe patterns of a transparent lens. The proposed method can measure both surfaces simultaneously without using any additional equipment. The measurement accuracy is comparable to that of precision profilometers with higher resolution and efficiency. Besides, the measuring range

of deflectometry can achieve a sub-millimeter level and the measuring system has better performance on the robustness against environmental disturbance and ease of operation. It is especially suited to the in-situ measurement of optical lenses at the early stage of polishing. As a consequence, this method is of great significance for the precision manufacturing of aspheric and freeform lenses.

Declaration of Competing Interest

The authors claim that there is no conflict of interest concerning this manuscript.

CRediT authorship contribution statement

Junqiang Ye: Investigation, Methodology, Writing - original draft. **Zhenqi Niu:** Data curation, Investigation. **Xiangchao Zhang:** Supervision, Writing - review & editing, Conceptualization, Funding acquisition. **Wei Wang:** Validation. **Min Xu:** Supervision.

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